

Sparse Ensembles: Do Routing Mechanisms Benefit from Ensemble Effects?

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Abstract

Ensemble effects among predictive models have been studied extensively and are a well-established source of improved generalization. In sparse ensembles, however, prediction is only one part of the problem: inputs must also be routed to the appropriate models. Whether this routing mechanism itself benefits from ensemble effects remains largely unexplored. We provide a systematic study of ensembling the routing mechanism, examining when increasing the number of routers improves performance, how their outputs should be combined, and whether explicitly encouraging diversity among routers is beneficial. Across classification and regression tasks, we characterize how these choices interact with the size and capacity of the underlying models and the routing ensemble.

1 Introduction

Ensemble methods are a well-established class of machine learning techniques that improve generalization by aggregating the predictions of multiple models. Their effectiveness is typically attributed to two complementary factors: the performance of the individual predictors and the diversity among them. Improving individual predictors reduces bias, while diversity promotes error de-correlation, allowing aggregation to cancel out model-specific mistakes.

Classical ensemble methods induce diversity implicitly via bagging, such as random forests, where predictors are trained on different subsets of the data, whereas boosting methods, such as LightGBM, construct predictors sequentially to focus on errors made by earlier models. In deep learning, diversity can be encouraged explicitly through mechanisms such as negative-correlation learning (Chen & Yao, 2009) or expert orthogonalization (Liu et al., 2024). However, Fort et al. (2019) showed that, in many cases, initializing experts with different random weights is sufficient to induce substantial diversity.

Sparse ensembles, in which only a subset of predictors is active for a given input, have gained prominence because of their computational advantages in large-scale language modeling. In the most prominent setting, each expert, or subset of experts, is responsible for a particular region of the input domain, allowing the ensemble to approximate the capacity of a larger model while retaining the computational cost of a smaller one. Existing variants balance computation and ensemble effects in different ways: selecting the top- k predictors for each input (Shazeer et al., 2017), maximizing cost reduction by routing each input to a single predictor (Fedus et al., 2022), or allocating a dynamic number of predictors per input (Wang et al., 2025). Routing-based methods decompose prediction into two coupled subproblems: a *delegation problem*, handled by a *router* or *delegator*, and a *prediction problem*, handled by the *experts* or *predictors*.

While ensembles typically focus on improving the predictive capacity of the experts, the delegation mechanism itself may also benefit from ensembling. This motivates studying whether increasing the number of delegators improves the allocation of unseen samples to predictors and, consequently, the overall performance of the ensemble. Despite recent methods that combine multiple routers (Zhang et al., 2025; Ran et al., 2025), the effect of scaling the number of delegators remains underexplored. In particular, to our knowledge, there is no systematic analysis of this setting through the lens of ensemble theory.

Contributions In this work we provide a systematic study of scaling the number of delegators in image classification tasks, focusing on CIFAR10 (Krizhevsky, 2009) and SVHN (Netzer et al., 2011), and tabular regression tasks, Bike sharing dataset (Fanaee-T, 2013) and Energy production dataset (Zaki, 2026). Specifically this work systematically tests scaling of number of delegators, number of predictors, delegator widths, and predictor widths. Furthermore, two aggregation methods are tested, probabilities product and probabilities sum. We find:

- **(C1) When scaling the number of delegators helps:** We identify the model-capacity regimes in which increasing the number of delegators improves ensemble performance.
- **(C2) Predictor ambiguity:** We propose a modification to the learning scheme in which delegators learn from both predictor performance and predictor ambiguity, while predictor ambiguity gradients are stopped from influencing the predictors.
- **(C3) Delegator ambiguity:** We propose a potential new performance–ambiguity decomposition for delegators and empirically validate whether delegators also benefit from the removal of the ambiguity gradient.
- **(C4) Delegator aggregation:** We empirically compare arithmetic and normalized geometric aggregation to determine which rule works better for combining delegator weights under joint training.

2 Background

We consider a supervised learning setting with a training dataset $\mathbb{D} = \{(\mathbf{x}_n, \mathbf{y}_n)\}_{n=1}^N$ of N i.i.d. samples, where $\mathbf{x} \in \mathbb{R}^{d_x}$ denotes the input and $\mathbf{y} \in \mathbb{R}^{d_y}$ the corresponding real-valued output for regression, or $\mathbf{y} \in \{0, 1\}^{d_y}$ a one-hot target for classification. We write $\Delta_M = \{\mathbf{a} \in \mathbb{R}^M : a_m \geq 0, \sum_{m=1}^M a_m = 1\}$ for the probability simplex.

We define an ensemble of M predictors $f_m = f_{\theta_m}$, with parameters $\Theta = (\theta_1, \dots, \theta_M)$, and an ensemble of L delegators $g_l = g_{\phi_l}$, with parameters $\Phi = (\phi_1, \dots, \phi_L)$. A predictor produces an output vector $\hat{\mathbf{y}}_m(\mathbf{x}) = f_m(\mathbf{x})$, which is a probability vector for classification. A delegator produces a weight proposal $\mathbf{a}_l(\mathbf{x}) = g_l(\mathbf{x}) \in \Delta_M$. Thus, predictors estimate the target, while delegators determine how predictors are combined.

For any weights $\mathbf{a} \in \Delta_M$, define the predictor aggregate as prescribed in Wood et al. (2023):

$$F(\mathbf{x}; \mathbf{a}) = \begin{cases} \sum_{m=1}^M a_m \hat{\mathbf{y}}_m(\mathbf{x}), & \text{regression,} \\ \frac{\odot_{m=1}^M \hat{\mathbf{y}}_m(\mathbf{x})^{a_m}}{\sum_{c=1}^{d_y} \prod_{m=1}^M \hat{y}_{m,c}(\mathbf{x})^{a_m}}, & \text{classification.} \end{cases} \quad (1)$$

Here $\odot$ denotes componentwise multiplication and $\mathbf{v}^{\odot \mathbf{a}}$ denotes componentwise exponentiation. The delegator proposals are combined into $\bar{\mathbf{a}}(\mathbf{x}) \in \Delta_M$ using either an arithmetic or normalized geometric mean (Section 4.4). For an input $\mathbf{x}$, the final prediction is $\hat{\mathbf{y}} = F(\mathbf{x}; \bar{\mathbf{a}}(\mathbf{x}))$, and the prediction induced by delegator l alone is $\hat{\mathbf{y}}^{(l)} = F(\mathbf{x}; \mathbf{a}_l(\mathbf{x}))$. With no delegators ($L = 0$), we use uniform weights $\bar{a}_m = 1/M$.

2.1 Predictor performance and ambiguity

For a fixed sample $(\mathbf{x}, \mathbf{y})$, write $\hat{\mathbf{y}}_m = f_m(\mathbf{x})$ for each predictor’s output. Let $\text{SE}(\hat{\mathbf{y}}, \mathbf{y}) = d_y^{-1} \|\hat{\mathbf{y}} - \mathbf{y}\|_2^2$ and $\text{CE}(\hat{\mathbf{y}}, \mathbf{y}) = -\sum_{c=1}^{d_y} y_c \log \hat{y}_c$ denote squared error and cross-entropy. For any weights $\mathbf{a} \in \Delta_M$, the classical squared-error decomposition (Krogh & Vedelsby, 1994) is

$$\text{SE}(F(\mathbf{x}; \mathbf{a}), \mathbf{y}) = \underbrace{\sum_{m=1}^M a_m \text{SE}(\hat{\mathbf{y}}_m, \mathbf{y})}_{\text{weighted predictor loss } \mathcal{E}_P(\mathbf{a})} - \underbrace{\sum_{m=1}^M a_m \text{SE}(F(\mathbf{x}; \mathbf{a}), \hat{\mathbf{y}}_m)}_{\text{weighted predictor ambiguity } \mathcal{A}_P(\mathbf{a})}. \quad (2)$$

For cross-entropy with geometric aggregation, the corresponding decomposition is (Wood et al., 2023)

$$\text{CE}(F(\mathbf{x}; \mathbf{a}), \mathbf{y}) = \underbrace{\sum_{m=1}^M a_m \text{CE}(\hat{\mathbf{y}}_m, \mathbf{y})}_{\text{weighted predictor loss } \mathcal{E}_P(\mathbf{a})} - \underbrace{\sum_{m=1}^M a_m \text{KL}(F(\mathbf{x}; \mathbf{a}) \parallel \hat{\mathbf{y}}_m)}_{\text{weighted predictor ambiguity } \mathcal{A}_P(\mathbf{a})}. \quad (3)$$

In both cases, $\mathcal{E}_P$ measures individual predictor accuracy, while $\mathcal{A}_P$ measures disagreement with the aggregate output. Lower individual loss and greater ambiguity at fixed individual loss improve ensemble performance. We use ℓ for the task loss (SE or CE) and d for its ambiguity measure (SE or KL), so both identities read $\ell(F(\mathbf{x}; \mathbf{a}), \mathbf{y}) = \mathcal{E}_P(\mathbf{a}) - \mathcal{A}_P(\mathbf{a})$.

Explicit maximization of ambiguity has been shown to be potentially harmful, since it can increase the bias and variance of the individual models, leading to higher generalization error (Ortega et al., 2022; Wood et al., 2023). In contrast, prior work indicates that using different random initializations is often sufficient to induce test-time diversity among ensemble members (Fort et al., 2019; Ortega et al., 2022). This motivates controlling ambiguity gradients separately from the loss decomposition itself (Section 4.2).

3 Delegator decompositions

3.1 Delegator performance and ambiguity

There is no target weight vector for a delegator. Instead, we assess each proposal through the predictor ensemble it induces. The loss $\ell(\hat{\mathbf{y}}^{(l)}, \mathbf{y})$ measures whether delegator l proposes weights that yield an accurate prediction. Its ambiguity $d(\hat{\mathbf{y}}, \hat{\mathbf{y}}^{(l)})$ measures disagreement with the final prediction, rather than disagreement between weight vectors themselves.

For arithmetic aggregation $\bar{\mathbf{a}} = L^{-1} \sum_{l=1}^L \mathbf{a}_l$, the final prediction is also the appropriate equal-weight aggregate of the induced predictions: their arithmetic mean for regression and their normalized geometric mean for classification. Applying the classical ambiguity decomposition (Equations 2–3) to these predictions gives

$$\ell(\hat{\mathbf{y}}, \mathbf{y}) = \underbrace{\frac{1}{L} \sum_{l=1}^L \ell(\hat{\mathbf{y}}^{(l)}, \mathbf{y})}_{\text{individual delegator loss } \mathcal{E}_D} - \underbrace{\frac{1}{L} \sum_{l=1}^L d(\hat{\mathbf{y}}, \hat{\mathbf{y}}^{(l)})}_{\text{delegator ambiguity } \mathcal{A}_D}. \quad (4)$$

The derivation for both tasks is given in Appendix A.2. Thus, each delegator should propose weights yielding a good predictor ensemble, while diversity is measured in the outputs these proposals induce. A larger $\mathcal{A}_D$ is beneficial at fixed $\mathcal{E}_D$. Each delegator’s loss also has the predictor decomposition

$$\ell(\hat{\mathbf{y}}^{(l)}, \mathbf{y}) = \mathcal{E}_P(\mathbf{a}_l) - \mathcal{A}_P(\mathbf{a}_l). \quad (5)$$

Equation 4 does not generally hold for geometric aggregation of delegation weights. The two terms remain defined, but their difference need not equal the final loss. Both terms require $L \geq 1$.

3.2 Delegator error

We also separate the final loss into a predictor loss under reference weights and a delegation regret. This is distinct from the performance–ambiguity decompositions above. For fixed predictor outputs and a sample $(\mathbf{x}, \mathbf{y})$, define the unrestricted oracle weights by

$$\mathbf{a}^*(\mathbf{x}, \mathbf{y}) \in \arg \min_{\mathbf{a} \in \Delta_M} \ell(F(\mathbf{x}; \mathbf{a}), \mathbf{y}). \quad (6)$$

The loss decomposes exactly as

$$\ell(\hat{\mathbf{y}}, \mathbf{y}) = \underbrace{\ell(F(\mathbf{x}; \mathbf{a}^*), \mathbf{y})}_{\text{predictor loss under optimal weights}} + \underbrace{[\ell(\hat{\mathbf{y}}, \mathbf{y}) - \ell(F(\mathbf{x}; \mathbf{a}^*), \mathbf{y})]}_{\text{delegator regret}}. \quad (7)$$

The first term is the minimum loss attainable by any admissible delegation weights for the given predictor outputs; the second is nonnegative. For regression, the first term is the squared distance to the convex hull of $\{\hat{\mathbf{y}}_m\}_{m=1}^M$, divided by d_y , and is zero exactly when the target lies in that hull. This convex-hull interpretation does not apply to geometrically aggregated classification probabilities.

The unrestricted, label-dependent oracle can be overly optimistic: with sufficiently many predictors, arbitrary sample-specific weights may interpolate regression targets. We therefore use a restricted delegator oracle with the same architecture as the learned delegators. With predictors fixed, this oracle seeks weights that reduce the ensemble loss. Its aggregate weights are denoted by $\mathbf{a}^\dagger(\mathbf{x})$. They provide a reference for attributing the remaining loss between predictors and delegators, but need not be optimal. The fitting and data-splitting procedure is described in Appendix A.5.

Replacing $\mathbf{a}^*$ by $\mathbf{a}^\dagger$ gives

$$\ell(\hat{\mathbf{y}}, \mathbf{y}) = \underbrace{\ell(F(\mathbf{x}; \mathbf{a}^\dagger), \mathbf{y})}_{\text{predictor loss under restricted-oracle weights}} + \underbrace{[\ell(\hat{\mathbf{y}}, \mathbf{y}) - \ell(F(\mathbf{x}; \mathbf{a}^\dagger), \mathbf{y})]}_{\text{restricted delegator regret}}. \quad (8)$$

The second term measures the difference relative to the restricted oracle and may be negative, since the oracle is not sample-wise optimal. The first term is the predictor loss remaining under that reference. Together, they distinguish potential gains from better delegation from the loss that remains after delegation is improved.

4 Experiments

Our claims are validated empirically over selection of four datasets: CIFAR10 (Krizhevsky, 2009), SVHN (Netzer et al., 2011), Bike sharing dataset (Fanaee-T, 2013), and Energy production dataset (Zaki, 2026). For each dataset, the inputs $\mathbf{x}$ and targets $\mathbf{y}$ were normalized where applicable, and categorical integer-valued features were one-hot encoded. For each random seed, the data were independently partitioned into training and validation sets using an 85%–15% split. Each training run is repeated across 5 different seeds.

For the tabular regression tasks, Bike and Energy, we use fully connected MLPs with two hidden layers, and optimize them using the mean squared error loss, ℓ_{MSE} . The hidden-layer widths decrease progressively toward the output: the two-layer architecture uses widths $(2b, b)$, where b is a model-specific base width. This parameterization preserves a consistent architectural shape while allowing the capacities of the predictor and delegator networks to be varied independently through their respective base widths. Regression tasks are learned for 2000 epochs.

For the image-classification tasks, CIFAR-10 and SVHN, we use three-layer MLPs with hidden widths being $(3b, 2b, b)$, where b is a model-specific base width. Classification tasks are optimized using the cross-entropy loss, ℓ_{CE} . Remaining training hyper-parameters are specified in Table 1. Image tasks are learned for 100 epochs.

Following the load-balancing motivation of Shazeer et al. (2017) and Fedus et al. (2022), we penalize nonuniform predictor usage through normalized Gini impurity. For a minibatch $\mathbb{B} \subseteq \mathbb{D}$, the final training objective is

$$\begin{aligned} \mathcal{L}_{\mathbb{B}} &= \frac{1}{|\mathbb{B}|} \sum_{(\mathbf{x}, \mathbf{y}) \in \mathbb{B}} \ell_{\text{train}}(\mathbf{x}, \mathbf{y}) + \lambda \frac{\sum_{m=1}^M u_m^2 - 1/M}{1 - 1/M}, \\ \mathbf{u} &= \frac{1}{|\mathbb{B}|} \sum_{(\mathbf{x}, \mathbf{y}) \in \mathbb{B}} \bar{\mathbf{a}}(\mathbf{x}), \end{aligned} \quad (9)$$

where $M > 1$, $\lambda = 0.2$, and ℓ_{train} has the value of the ensemble loss but the selected training gradients defined in Section 4.2. This penalty encourages balanced predictor use across the minibatch.

How we analyze the experiments. For every experiment, we ask three separate questions:

1. **Did the full system get better?** We compare the validation loss of the complete predictor and delegator ensemble. This is the final performance that we care about.

2. **What happened to the individual ensemble members?** The predictor performance–ambiguity decomposition tells us how well the individual predictors perform and how different their predictions are. The delegator performance–ambiguity decomposition tells us the same for the individual delegators, using the predictions induced by their weights.
3. **Where does the remaining validation loss come from?** We split the total validation loss into the loss under restricted-oracle weights and the delegator regret using Equation 8. The restricted-oracle loss is what remains after we try to improve the delegator weights while keeping the predictors fixed. The regret is the difference between this loss and the actual loss. We report this regret as a percentage of the total validation loss to estimate how much of the remaining loss is the delegators’ fault.

4.1 Scaling the number of delegators

We next investigate whether delegation itself benefits from an ensemble effect, and under which model-capacity regimes such benefits emerge (Contribution **(C1)**). In particular, we study whether increasing the number of delegators is most useful when the predictor ensemble is small, the predictors have limited capacity, or the individual delegators are themselves relatively small.

To isolate the effect of the number of delegators, we vary $L \in \{0, 1, 2, 4, 8, 16, 32\}$ while exhaustively evaluating all combinations of the number of predictors $M \in \{2, 4, 8, 16, 32\}$, predictor base width $b_M \in \{4, 8, 16\}$, and delegator base width $b_L \in \{4, 8, 16\}$ across all four tasks. The case $L = 0$ corresponds to fixed uniform delegation weights $\bar{a}$, whereas $L = 1$ recovers a setting closely related to a standard mixture-of-experts model. Each experimental condition is evaluated over five independent random seeds, yielding 6300 training runs. We use the `on`[$\mathcal{A}_P$] setting (Section 4.2) and the `off`[$\mathcal{A}_D$] setting (Section 4.3).

4.2 Predictor ambiguity gradient to delegators

As discussed in Section 2, explicitly maximizing predictor ambiguity can harm individual predictor performance. We therefore stop the predictor ambiguity gradient from influencing the predictor parameters Θ in both conditions. The experiment only changes whether the delegators learn from predictor ambiguity $\mathcal{A}_P$:

$$\underbrace{\mathcal{E}_P(\bar{a}) - \mathcal{A}_P(\bar{a})}_{\text{on}[\mathcal{A}_P]} \quad \text{or} \quad \underbrace{\mathcal{E}_P(\bar{a})}_{\text{off}[\mathcal{A}_P]}. \quad (10)$$

In the `on`[$\mathcal{A}_P$] condition, delegators learn from both predictor accuracy and complementarity. In the `off`[$\mathcal{A}_P$] condition, they learn only from predictor performance. Predictors learn only from $\mathcal{E}_P$ in both conditions. The stop-gradient equations are given in Appendix A.4.

We compare `on`[$\mathcal{A}_P$] and `off`[$\mathcal{A}_P$] with `on`[$\mathcal{A}_D$] and arithmetic delegator aggregation, corresponding to Contribution **(C2)**. Following the protocol of the preceding experiment, we sample 100 configurations from the parameter sets in Table 2 and evaluate both conditions for each configuration and each of the four tasks, holding all other experimental factors fixed. This yields 800 experimental conditions. Each condition is evaluated over five independent random seeds, for a total of 4,000 training runs. The comparison in which predictor ambiguity also trains the predictors is reported separately in Appendix A.3.

Ensemble performance Figure 1 shows the relative validation performance difference between the two predictor ambiguity conditions.

Predictors’ performance

Delegators’ performance

Delegators’ regret analysis We found that in 85% cases the restricted oracle was able to improve the validation loss.

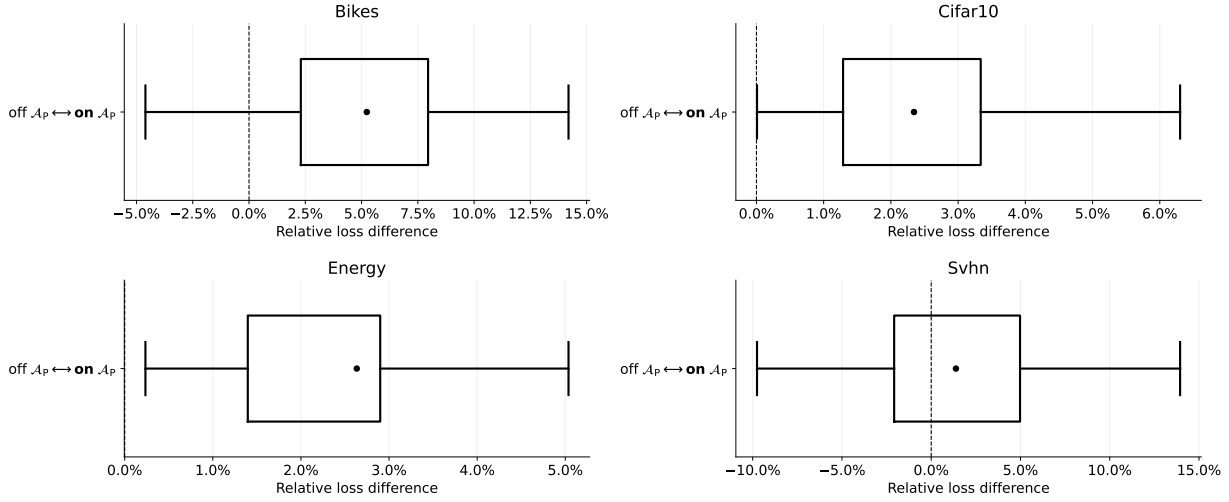


Figure 1: Predictor ambiguity gradient experiment: Distribution of the relative validation-loss differences between $\text{on}[\mathcal{A}_P]$ and $\text{off}[\mathcal{A}_P]$, computed on validation samples. The better condition is marked in bold on the axis. The black dot denotes the mean relative difference.

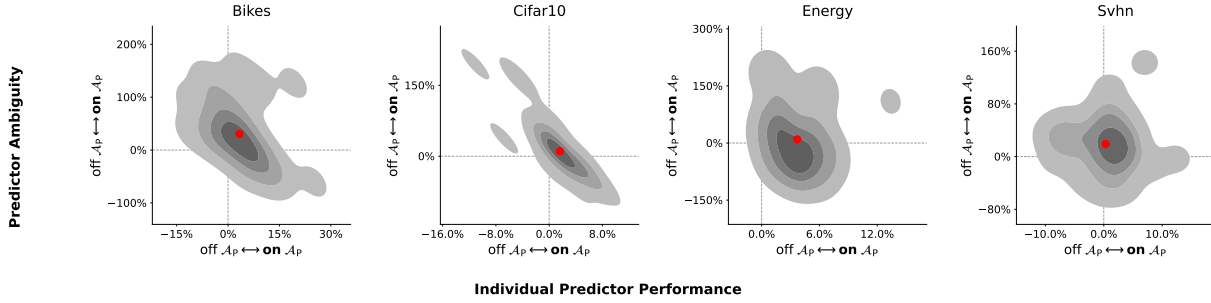


Figure 2: Predictor ambiguity gradient experiment: Distribution of the predictor performance differences (x-axis) and predictor ambiguity differences (y-axis) between $\text{on}[\mathcal{A}_P]$ and $\text{off}[\mathcal{A}_P]$, computed on validation samples. The better condition is marked in bold separately on each axis. The red dot denotes the two-dimensional mean.

4.3 Delegator ambiguity gradient to delegators

We next test whether delegators should learn from delegator ambiguity $\mathcal{A}_D$. We compare two conditions. In $\text{on}[\mathcal{A}_D]$, delegators learn from both their individual performance and their ambiguity. In $\text{off}[\mathcal{A}_D]$, the ambiguity gradient is blocked, so each delegator is trained to produce an accurate ensemble prediction on its own. The two objectives used to train the delegators are

$$\underbrace{\mathcal{E}_P(\bar{a}) - \mathcal{A}_P(\bar{a})}_{\text{on}[\mathcal{A}_D]} \quad \text{or} \quad \underbrace{\mathcal{E}_D}_{\text{off}[\mathcal{A}_D]}. \quad (11)$$

The first expression equals $\mathcal{E}_D - \mathcal{A}_D$ under the $\text{on}[\mathcal{A}_P]$ condition used in this experiment. All other task-loss gradients to delegators are stopped; the load-balancing penalty remains active in both conditions. We enumerate the full stop-gradient equations in Appendix A.4.

To isolate the effect of delegator ambiguity for Contribution **(C3)**, we keep predictor ambiguity fixed at $\text{on}[\mathcal{A}_P]$ and use arithmetic delegator aggregation. We sample 100 configurations per task from the parameter sets in Table 2. For each configuration, we compare a matched pair of $\text{on}[\mathcal{A}_D]$ and $\text{off}[\mathcal{A}_D]$, holding all other

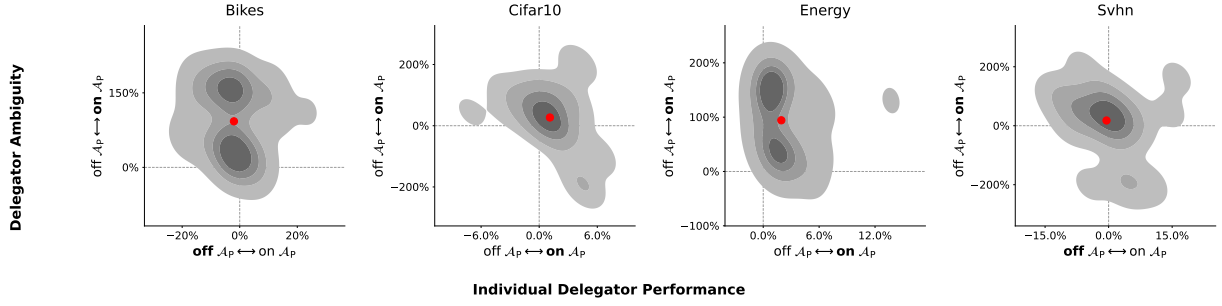


Figure 3: Predictor ambiguity gradient experiment: Distribution of the delegator performance differences (x-axis) and delegator ambiguity differences (y-axis) between $\text{on}[A_P]$ and $\text{off}[A_P]$, computed on validation samples. The better condition is marked in bold separately on each axis. The red dot denotes the two-dimensional mean.

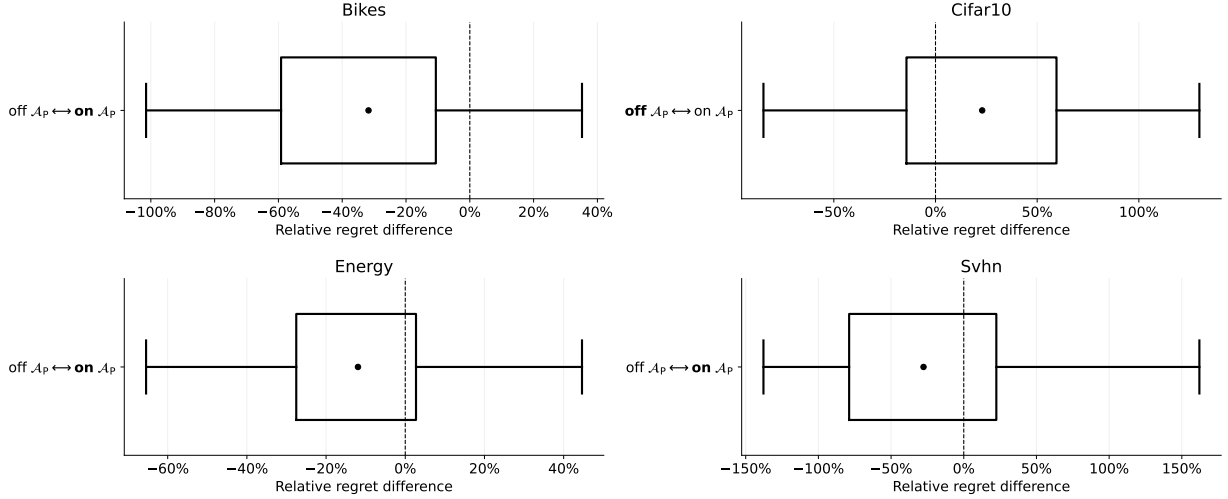


Figure 4: Predictor ambiguity gradient experiment: Distribution of the relative delegator-regret differences between $\text{on}[A_P]$ and $\text{off}[A_P]$, computed on validation samples. The bold condition has more delegator regret, meaning that more of its remaining loss is attributable to the delegators. The black dot denotes the mean relative difference.

factors fixed. Each condition is evaluated over five independent random seeds. Across the four tasks, this yields 800 experimental conditions and 4,000 training runs.

Ensemble performance

Predictors' performance

Delegators' performance

Delegators' regret analysis

4.4 Delegator's aggregation method

Previous work (Ortega et al., 2022; Wood et al., 2023) characterizes aggregation rules for predictors under a fixed set of weights α . In our setting, predictor and delegator parameters are optimized jointly, so the

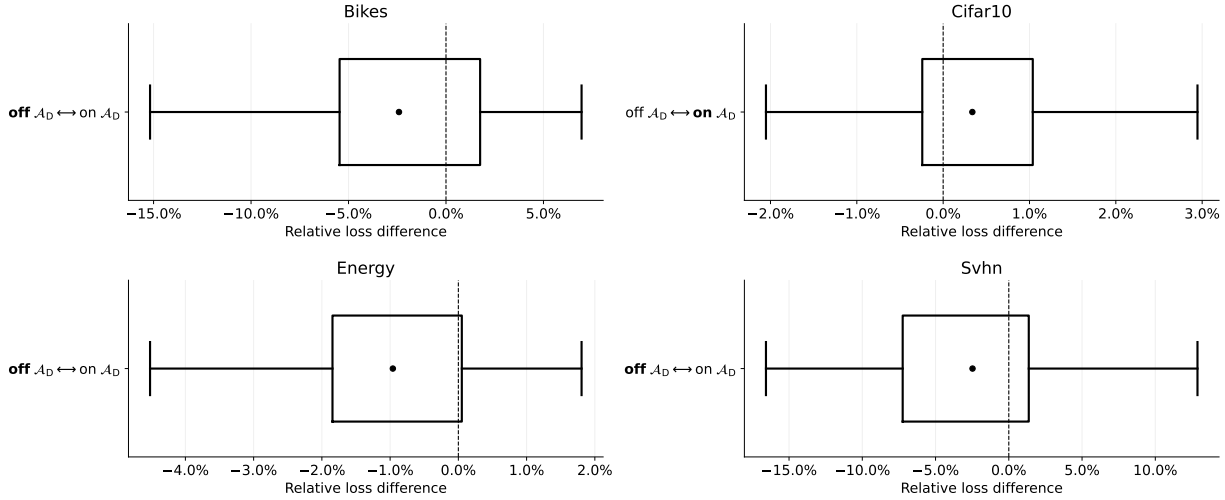


Figure 5: Delegator ambiguity gradient experiment: Distribution of the relative validation-loss differences between $\text{on}[\mathcal{A}_D]$ and $\text{off}[\mathcal{A}_D]$, computed on validation samples. The better condition is marked in bold on the axis. The black dot denotes the mean relative difference.

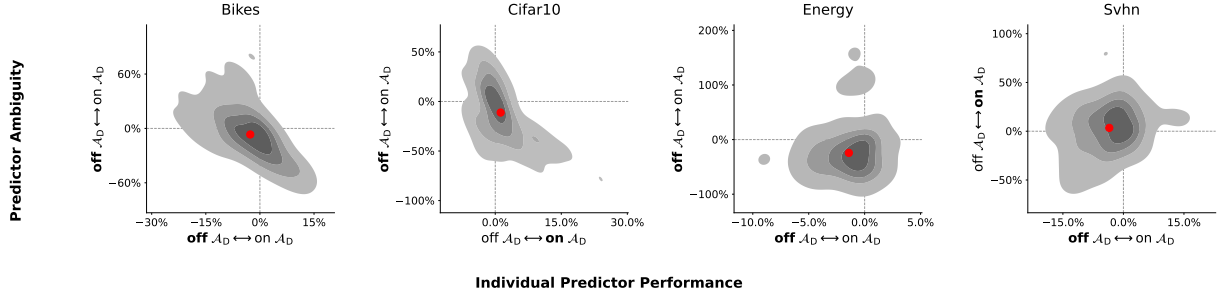


Figure 6: Delegator ambiguity gradient experiment: Distribution of the predictor performance differences (x-axis) and predictor ambiguity differences (y-axis) between $\text{on}[\mathcal{A}_D]$ and $\text{off}[\mathcal{A}_D]$, computed on validation samples. The better condition is marked in bold separately on each axis. The red dot denotes the two-dimensional mean.

aggregation rule also affects how the two ensembles co-adapt during training. We therefore compare the two aggregation rules empirically, corresponding to Contribution (C4).

We compare two aggregation rules over the L delegators: the arithmetic mean, $\bar{\mathbf{a}}(\mathbf{x}) = L^{-1} \sum_{l=1}^L \mathbf{a}_l(\mathbf{x})$, and the normalized geometric mean, $\bar{\mathbf{a}}(\mathbf{x}) = Z^{-1} \odot_{l=1}^L \mathbf{a}_l(\mathbf{x})^{L^{-1}}$, where Z normalizes the weights to sum to one. The predictor aggregation in Equation 1 remains unchanged. We use the $\text{off}[\mathcal{A}_P]$ setting (Section 4.2) and the $\text{off}[\mathcal{A}_D]$ setting (Section 4.3).

To isolate the effect of the aggregation rule, we construct 100 experimental configurations by sampling the remaining experimental variables from the sets reported in Table 2. For each configuration and each of the four tasks, we evaluate a matched pair of experiments differing only in the delegator aggregation rule: arithmetic mean or geometric mean. This yields 800 experimental conditions. Each condition is evaluated over five independent random seeds, for a total of 4,000 training runs.

Ensemble performance Figure 9 shows that arithmetic aggregation has lower mean validation loss on Bikes, CIFAR-10, and Energy. The difference is largest on Bikes and smaller on CIFAR-10 and Energy. On SVHN, the mean difference is close to zero and the distribution is broad and spans both directions, so neither

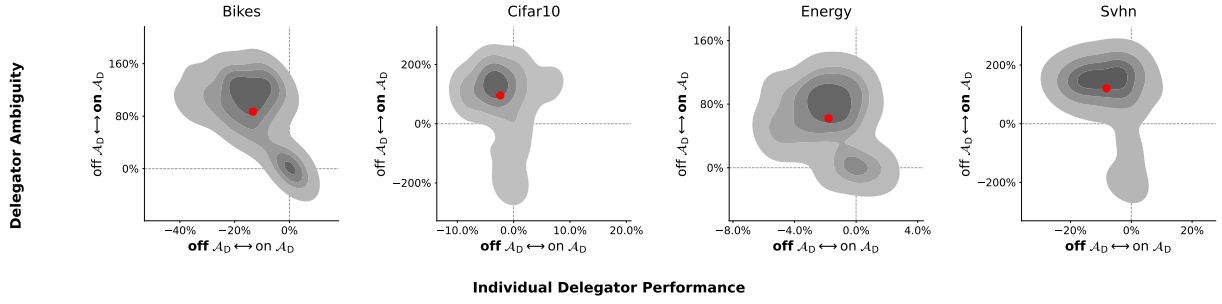


Figure 7: Delegator ambiguity gradient experiment: Distribution of the delegator performance differences (x-axis) and delegator ambiguity differences (y-axis) between $\text{on}[\mathcal{A}_D]$ and $\text{off}[\mathcal{A}_D]$, computed on validation samples. The better condition is marked in bold separately on each axis. The red dot denotes the two-dimensional mean.

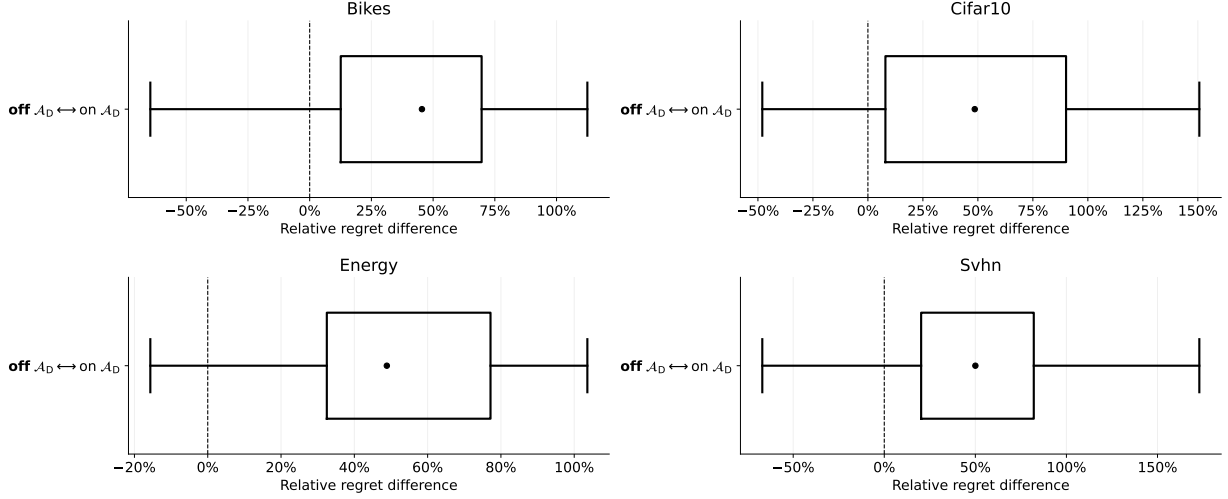


Figure 8: Delegator ambiguity gradient experiment: Distribution of the relative delegator-regret differences between $\text{on}[\mathcal{A}_D]$ and $\text{off}[\mathcal{A}_D]$, computed on validation samples. The bold condition has more delegator regret, meaning that more of its remaining loss is attributable to the delegators. The black dot denotes the mean relative difference.

aggregation rule has a clear average advantage. Thus, arithmetic aggregation performs better on three of the four tasks, although the average improvement is modest outside Bikes.

Predictors’ performance Figure 10 shows a consistent trade-off. Arithmetic aggregation produces substantially greater predictor ambiguity on all four tasks, whereas geometric aggregation produces slightly better individual predictor performance. On Bikes, CIFAR-10, and Energy, the increased predictor ambiguity under arithmetic aggregation accompanies the lower overall validation loss reported above. On SVHN, the ambiguity difference is smaller and the predictor-performance distribution is much wider, which is consistent with the mixed overall result on that task.

Delegators’ performance Figure 11 shows the opposite trade-off for the delegators. Geometric aggregation produces substantially greater delegator ambiguity on every task, whereas arithmetic aggregation produces better individual delegator performance. The delegator-ambiguity differences are tightly concentrated, whereas the individual-performance differences vary more across configurations.

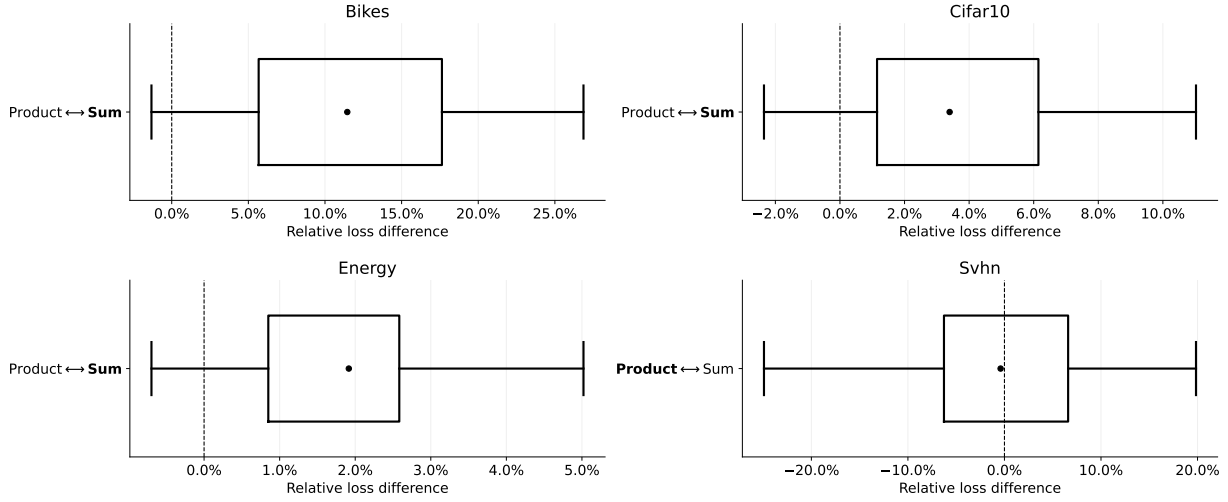


Figure 9: Delegator aggregation experiment: Distribution of the relative validation-loss differences between arithmetic and normalized geometric aggregation, computed on validation samples. The better aggregation rule is marked in bold on the axis. The black dot denotes the mean relative difference.

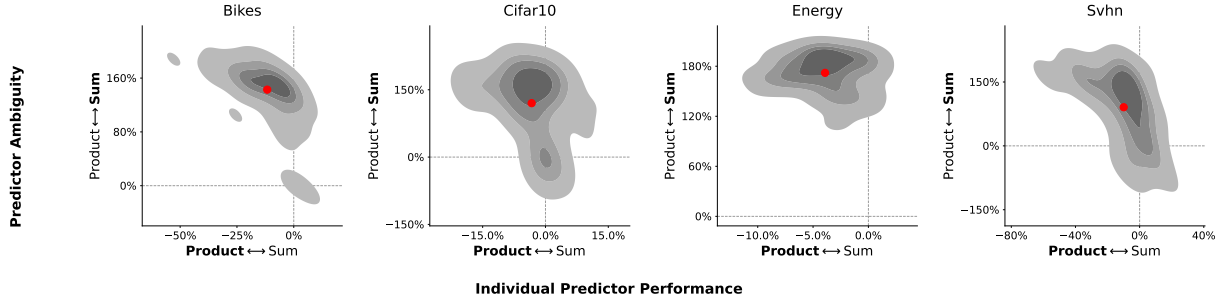


Figure 10: Delegator aggregation experiment: Distribution of the predictor performance differences (x-axis) and predictor ambiguity differences (y-axis) between arithmetic and normalized geometric aggregation, computed on validation samples. The better aggregation rule is marked in bold separately on each axis. The red dot denotes the two-dimensional mean.

Delegators’ regret analysis Figure 12 shows greater delegator regret under geometric aggregation on every task, with the interquartile range remaining positive in all four cases. Therefore, under geometric aggregation, a larger share of the remaining validation loss is attributable to the delegators rather than the predictors. The additional delegator ambiguity produced by geometric aggregation does not result in better delegation.

Taken together, the aggregation experiment favors the arithmetic mean as the stronger default for combining delegator weights. It produces lower mean validation loss on three tasks and comparable mean performance on SVHN. Although geometric aggregation produces greater delegator ambiguity, its higher delegator regret shows that this does not lead to better delegation.

4.5 Analysis

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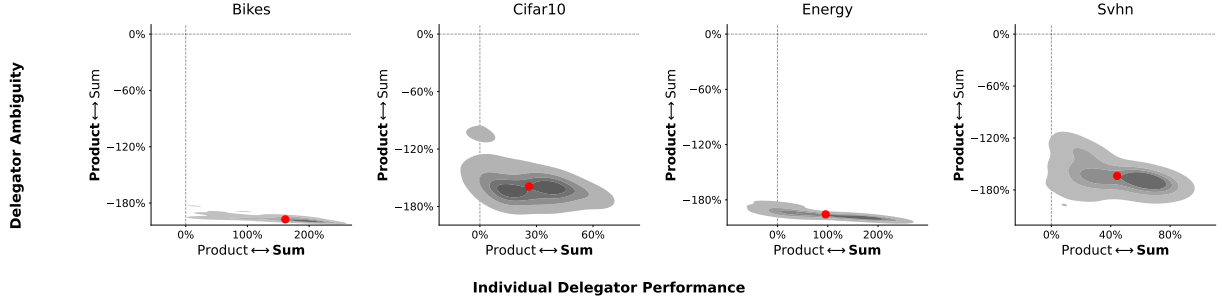


Figure 11: Delegator aggregation experiment: Distribution of the delegator performance differences (x-axis) and delegator ambiguity differences (y-axis) between arithmetic and normalized geometric aggregation, computed on validation samples. The better aggregation rule is marked in bold separately on each axis. The red dot denotes the two-dimensional mean.

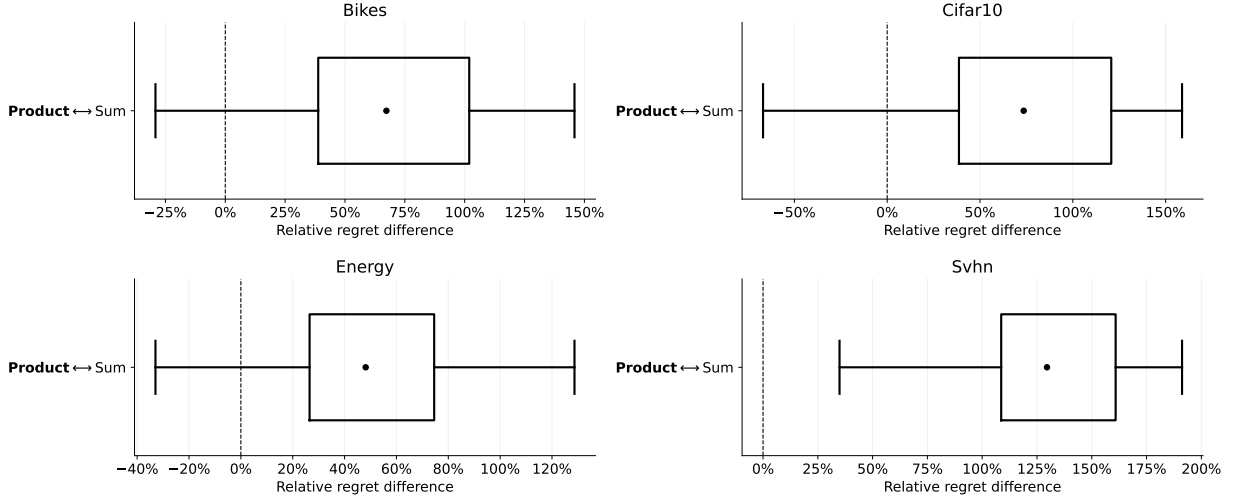


Figure 12: Delegator aggregation experiment: Distribution of the relative delegator-regret differences between arithmetic and normalized geometric aggregation, computed on validation samples. The bold aggregation rule has more delegator regret, meaning that more of its remaining loss is attributable to the delegators. The black dot denotes the mean relative difference.

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Table 1: Training and initialization hyperparameters used across experiments.

HYPERPARAMETER	VALUE
Kernel initialization	LeCun normal
Bias initialization	Zero
Hidden-layer activation	ReLU
Optimizer	Adam
Learning rate	10^{-3}
Weight-decay coefficient	10^{-4}
Tabular-task batch size	256
Image-task batch size	64
Tabular-task training epochs	2000
Image-task training epochs	50

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Table 2: Experimental values and their permitted set.

VARIABLE	SET	NOTES
Number of predictors (M)	$\{2, 4, 8, 16, 32\}$	
Number of delegators (L)	$\{0, 1, 2, 4, 8, 16, 32\}$	$L = 0$: uniform weights; $L = 1$: MoE.
Predictors' base width (b_M)	$\{4, 8, 16\}$	Predictor hidden layers are either $(2b, b)$ or $(3b, 2b)$.
Delegators' base width (b_L)	$\{4, 8, 16\}$	Delegator hidden layers are either $(2b, b)$ or $(3b, 2b)$.
Delegators' aggregation method	$\{\text{mean, geometric mean}\}$	See Section 4.4
Predictor ambiguity gradient to delegators	$\text{off}[\mathcal{A}_P], \text{on}[\mathcal{A}_P]$	Section 4.2.
Delegator ambiguity gradient to delegators	$\text{off}[\mathcal{A}_D], \text{on}[\mathcal{A}_D]$	Section 4.3.

A Appendix

A.1 Training specifications

A.2 Delegator performance–ambiguity decomposition under the sum rule

Fix a sample $(\mathbf{x}, \mathbf{y})$ and $L \geq 1$ delegators. Under the sum rule, their weight proposals are averaged:

$$\bar{\mathbf{a}} = \frac{1}{L} \sum_{l=1}^L \mathbf{a}_l, \quad \hat{\mathbf{y}}^{(l)} = F(\mathbf{x}; \mathbf{a}_l), \quad \hat{\mathbf{y}} = F(\mathbf{x}; \bar{\mathbf{a}}). \quad (12)$$

We show that this rule makes the final prediction the appropriate aggregate of the delegator-induced predictions, and then derive $\ell(\hat{\mathbf{y}}, \mathbf{y}) = \mathcal{E}_D - \mathcal{A}_D$.

Regression. Predictor outputs are combined by a weighted arithmetic mean. Writing $a_{l,m}$ for the weight assigned to predictor m by delegator l , we obtain

$$\begin{aligned} \hat{\mathbf{y}} &= \sum_{m=1}^M \bar{a}_m \hat{\mathbf{y}}_m = \frac{1}{L} \sum_{l=1}^L \sum_{m=1}^M a_{l,m} \hat{\mathbf{y}}_m = \frac{1}{L} \sum_{l=1}^L \hat{\mathbf{y}}^{(l)} \\ \mathbf{0} &= \frac{1}{L} \sum_{l=1}^L \hat{\mathbf{y}}^{(l)} - \hat{\mathbf{y}} = \sum_{l=1}^L (\hat{\mathbf{y}}^{(l)} - \hat{\mathbf{y}}) \end{aligned} \quad (13)$$

Expanding each individual squared error around $\hat{\mathbf{y}}$ gives

$$\begin{aligned} \mathcal{E}_D &= \frac{1}{Ld_y} \sum_{l=1}^L \left\| (\hat{\mathbf{y}}^{(l)} - \hat{\mathbf{y}}) + (\hat{\mathbf{y}} - \mathbf{y}) \right\|_2^2 \\ &= \frac{1}{Ld_y} \sum_{l=1}^L \left\| \hat{\mathbf{y}}^{(l)} - \hat{\mathbf{y}} \right\|_2^2 + \frac{2}{Ld_y} (\hat{\mathbf{y}} - \mathbf{y})^\top \underbrace{\sum_{l=1}^L (\hat{\mathbf{y}}^{(l)} - \hat{\mathbf{y}})}_{\mathbf{0}} + \frac{1}{Ld_y} \sum_{l=1}^L \left\| \hat{\mathbf{y}} - \mathbf{y} \right\|_2^2 \\ &= \mathcal{A}_D + SE(\hat{\mathbf{y}}, \mathbf{y}) \end{aligned} \quad (14)$$

The cross term vanishes, yielding

$$SE(\hat{\mathbf{y}}, \mathbf{y}) = \mathcal{E}_D - \mathcal{A}_D. \quad (15)$$

Classification. Predictor outputs are combined by a normalized weighted geometric mean. Assume positive predictor probabilities, so the logarithms below are defined. For class c , the prediction induced by delegator l is

$$\hat{y}_c^{(l)} = \frac{1}{Z_l} \prod_{m=1}^M \hat{y}_{m,c}^{a_{l,m}}, \quad Z_l = \sum_{c=1}^{d_y} \prod_{m=1}^M \hat{y}_{m,c}^{a_{l,m}}. \quad (16)$$

Taking the geometric mean across delegators gives

$$\prod_{l=1}^L (\hat{y}_c^{(l)})^{1/L} = \left(\prod_{l=1}^L Z_l^{-1/L} \right) \prod_{m=1}^M \hat{y}_{m,c}^{\frac{1}{L} \sum_{l=1}^L a_{l,m}} = \left(\prod_{l=1}^L Z_l^{-1/L} \right) \prod_{m=1}^M \hat{y}_{m,c}^{\bar{a}_m}. \quad (17)$$

The factor involving Z_l is the same for every class and cancels when the probabilities are normalized. Hence the final prediction is the normalized geometric mean of the induced predictions:

$$\hat{y}_c = \frac{1}{Z_D} \prod_{l=1}^L (\hat{y}_c^{(l)})^{1/L}, \quad Z_D = \sum_{c=1}^{d_y} \prod_{l=1}^L (\hat{y}_c^{(l)})^{1/L}. \quad (18)$$

Taking logarithms yields

$$\frac{1}{L} \sum_{l=1}^L \log \hat{y}_c^{(l)} = \log \hat{y}_c + \log Z_D. \quad (19)$$

Using $\sum_c y_c = 1$, the mean individual delegator loss is therefore

$$\begin{aligned} \mathcal{E}_D &= - \sum_{c=1}^{d_y} y_c \left(\frac{1}{L} \sum_{l=1}^L \log \hat{y}_c^{(l)} \right) \\ &= \text{CE}(\hat{\mathbf{y}}, \mathbf{y}) - \log Z_D. \end{aligned} \quad (20)$$

Similarly, using $\sum_c \hat{y}_c = 1$, the mean delegator ambiguity is

$$\begin{aligned} \mathcal{A}_D &= \frac{1}{L} \sum_{l=1}^L \text{KL}(\hat{\mathbf{y}} \| \hat{\mathbf{y}}^{(l)}) \\ &= \sum_{c=1}^{d_y} \hat{y}_c \left(\log \hat{y}_c - \frac{1}{L} \sum_{l=1}^L \log \hat{y}_c^{(l)} \right) = -\log Z_D. \end{aligned} \quad (21)$$

Subtracting the two expressions gives

$$\text{CE}(\hat{\mathbf{y}}, \mathbf{y}) = \mathcal{E}_D - \mathcal{A}_D. \quad (22)$$

Equations 15 and 22 establish Equation 4. Both rely on arithmetic aggregation of delegation weights; geometric weight aggregation does not generally produce the required mean of the induced predictions.

A.3 Verification of ensemble theory

The main predictor ambiguity experiment compares `off` [$\mathcal{A}_P$] and `on` [$\mathcal{A}_P$]. In both conditions, predictor ambiguity does not update the predictors. Here we include the additional setting in which predictor ambiguity updates both predictors and delegators. We use the original names `None`, `Delegators`, and `Both` for this three-way comparison. Their definitions and their relation to the main-paper notation are shown in Table 3.

Table 3: Objectives used to update predictors and delegators in the three-way predictor ambiguity experiment. All terms use the learned aggregate weights $\bar{\mathbf{a}}$. Each column updates only its own parameter group.

Setting	Main notation	Updates predictors	Updates delegators
<code>Both</code>	appendix only	$\mathcal{E}_P - \mathcal{A}_P$	$\mathcal{E}_P - \mathcal{A}_P$
<code>Delegators</code>	<code>on</code> [$\mathcal{A}_P$]	$\mathcal{E}_P$	$\mathcal{E}_P - \mathcal{A}_P$
<code>None</code>	<code>off</code> [$\mathcal{A}_P$]	$\mathcal{E}_P$	$\mathcal{E}_P$

We compare all three settings with `on` [$\mathcal{A}_D$] and arithmetic delegator aggregation. We sample 100 configurations from the parameter sets in Table 2 and evaluate all three settings for each configuration and each of the four tasks, holding all other experimental factors fixed. This yields 1,200 experimental conditions. Each condition is evaluated over five independent random seeds, for a total of 6,000 training runs.

A.4 Gradient routing

We enumerate all combinations of predictor ambiguity gradient (PAG) and delegator ambiguity gradient (DAG) settings. These abbreviations are used only in this appendix. We assume arithmetic delegation aggregation and $L \geq 1$, so the ensemble loss satisfies

$$\ell = \mathcal{E}_P(\bar{\mathbf{a}}) - \mathcal{A}_P(\bar{\mathbf{a}}) = \mathcal{E}_D - \mathcal{A}_D, \quad \mathcal{E}_D = \frac{1}{L} \sum_{l=1}^L [\mathcal{E}_P(\mathbf{a}_l) - \mathcal{A}_P(\mathbf{a}_l)]. \quad (23)$$

The operator sg_{Θ} preserves its argument's value but stops its gradient with respect to predictor parameters. Similarly, sg_{Φ} stops gradients with respect to delegator parameters, while sg stops all gradients. The expressions below define ℓ_{train} : each has the same numerical value as ℓ , but supplies the gradients prescribed by its setting.

All settings also include the auxiliary losses, including the load-balancing penalty in Equation 9. These are added unchanged and are not subject to the stop-gradient operations below.

PAG = Both, DAG = True Both predictors and delegators receive the full ensemble-loss gradient:

$$\ell_{\text{train}} = \ell. \quad (24)$$

PAG = Delegators, DAG = True Predictor ambiguity updates only delegators:

$$\ell_{\text{train}} = \mathcal{E}_P(\bar{\mathbf{a}}) - \text{sg}_{\Theta} [\mathcal{A}_P(\bar{\mathbf{a}})]. \quad (25)$$

PAG = None, DAG = True Predictor ambiguity contributes no gradient:

$$\ell_{\text{train}} = \mathcal{E}_P(\bar{\mathbf{a}}) - \text{sg} [\mathcal{A}_P(\bar{\mathbf{a}})]. \quad (26)$$

PAG = Both, DAG = False Predictors receive the full ensemble-loss gradient, while delegators learn only from their individual performance:

$$\begin{aligned} \ell_{\text{train}} = & \text{sg}_{\Phi} [\ell] \\ & + \text{sg}_{\Theta} [\mathcal{E}_D - \text{sg}_{\Phi} [\mathcal{A}_D] + \text{sg}_{\Phi} [\mathcal{A}_D - \mathcal{E}_D]]. \end{aligned} \quad (27)$$

PAG = Delegators, DAG = False Predictors learn only from their weighted individual performance. Delegators learn from their individual performance, which still accounts for predictor ambiguity:

$$\begin{aligned} \ell_{\text{train}} = & \text{sg}_{\Phi} [\mathcal{E}_P(\bar{\mathbf{a}}) - \text{sg}_{\Theta} [\mathcal{A}_P(\bar{\mathbf{a}})]] \\ & + \text{sg}_{\Theta} [\mathcal{E}_D - \text{sg}_{\Phi} [\mathcal{A}_D] + \text{sg}_{\Phi} [\mathcal{A}_D - \mathcal{E}_D]]. \end{aligned} \quad (28)$$

PAG = None, DAG = False Predictors learn only from their weighted individual performance. Delegators also ignore predictor ambiguity within each individual weight proposal:

$$\begin{aligned} \ell_{\text{train}} = & \text{sg}_{\Phi} [\mathcal{E}_P(\bar{\mathbf{a}}) - \text{sg} [\mathcal{A}_P(\bar{\mathbf{a}})]] \\ & + \text{sg}_{\Theta} \left[\frac{1}{L} \sum_{l=1}^L (\mathcal{E}_P(\mathbf{a}_l) - \text{sg}_{\Phi} [\mathcal{A}_P(\mathbf{a}_l)]) - \text{sg}_{\Phi} [\mathcal{A}_D] + \text{sg}_{\Phi} [\mathcal{A}_D - \mathcal{E}_D] \right]. \end{aligned} \quad (29)$$

In the DAG = False settings, the second bracket has numerical value zero but supplies the required delegator gradients. The correction term $\text{sg}_{\Phi} [\mathcal{A}_D - \mathcal{E}_D]$ therefore preserves the reported ensemble loss without changing either parameter group's updates.

A.5 Restricted delegator oracle training procedure

For candidate delegator parameters Φ' , let $\bar{\mathbf{a}}_{\Phi'}(\mathbf{x})$ denote their aggregate weights. With predictors fixed, the oracle minimizes

$$\mathcal{L}_{\text{oracle}}(\Phi') = \frac{1}{|\mathbb{D}_{\text{fit}}|} \sum_{(\mathbf{x}, \mathbf{y}) \in \mathbb{D}_{\text{fit}}} \ell(F(\mathbf{x}; \bar{\mathbf{a}}_{\Phi'}(\mathbf{x})), \mathbf{y}). \quad (30)$$

For each trained seed, we freeze all predictor parameters and cache the validation predictions and each delegator’s penultimate-layer features. Starting from the learned delegators, we fit only the kernels and biases of their final affine layers using full-batch L-BFGS with history size 10. The objective is Equation 30 with $\mathbb{D}_{\text{fit}} = \mathbb{D}_{\text{val}}$, without a routing tether, load balancing, weight decay, or ambiguity-gradient restrictions. We perform at most 100 updates and stop when the gradient norm is at most 10^{-6} or a nonfinite value is encountered. Each seed retains the weights with the lowest validation loss, including the unmodified initialization. The softmax routing makes this optimization generally nonconvex, so the resulting weights are a restricted reference rather than a certified optimum. Because validation labels participate in fitting and selection, this procedure quantifies recoverable loss on that dataset only.